MA1905 Operations with Binary and Hexadecimal Numbers

Show all carries, borrowing, and partial products. Give answers in the same number base as the question. Do not convert to decimal as your main method.

Add the following binary numbers.

1.

$$\begin{array}{r} 1011 \\ +\ 110 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 1101 \\ +\ 1011 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 10101 \\ +\ 1110 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 11101 \\ +\ 10111 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 101101 \\ +\ 11011 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 111011 \\ +\ 101101 \\ \hline \end{array}$$

Subtract the following binary numbers.

7.

$$\begin{array}{r} 1101 \\ -\ 101 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 10110 \\ -\ 1001 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 11001 \\ -\ 1011 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 100000 \\ -\ 1101 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 101101 \\ -\ 1110 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 110010 \\ -\ 10111 \\ \hline \end{array}$$

Multiply the following binary numbers. Show all partial products.

13.

$$\begin{array}{r} 101 \\ \times\ 11 \\ \hline \end{array}$$

14.

$$\begin{array}{r} 110 \\ \times\ 101 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 1011 \\ \times\ 11 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 1101 \\ \times\ 101 \\ \hline \end{array}$$

17.

$$\begin{array}{r} 1010 \\ \times\ 110 \\ \hline \end{array}$$

18.

$$\begin{array}{r} 1110 \\ \times\ 1011 \\ \hline \end{array}$$

Add the following hexadecimal numbers.

19.

$$\begin{array}{r} 2A \\ +\ 17 \\ \hline \end{array}$$

20.

$$\begin{array}{r} 3F \\ +\ 28 \\ \hline \end{array}$$

21.

$$\begin{array}{r} 7B \\ +\ 4D \\ \hline \end{array}$$

22.

$$\begin{array}{r} A6 \\ +\ 39 \\ \hline \end{array}$$

23.

$$\begin{array}{r} 1E8 \\ +\ 2B7 \\ \hline \end{array}$$

24.

$$\begin{array}{r} 9AF \\ +\ 6D5 \\ \hline \end{array}$$

Subtract the following hexadecimal numbers.

25.

$$\begin{array}{r} 5C \\ -\ 27 \\ \hline \end{array}$$

26.

$$\begin{array}{r} A4 \\ -\ 3B \\ \hline \end{array}$$

27.

$$\begin{array}{r} F2 \\ -\ 6D \\ \hline \end{array}$$

28.

$$\begin{array}{r} 10A \\ -\ 7F \\ \hline \end{array}$$

29.

$$\begin{array}{r} 3B5 \\ -\ 1C8 \\ \hline \end{array}$$

30.

$$\begin{array}{r} A20 \\ -\ 4E9 \\ \hline \end{array}$$

Answer Key

Binary Addition (1–6)

Q1	Q2	Q3	Q4	Q5	Q6
10001_2	11000_2	100011_2	110100_2	1001000_2	1101000_2

Binary Subtraction (7–12)

Q7	Q8	Q9	Q10	Q11	Q12
1000_2	1101_2	1110_2	10011_2	11111_2	11011_2

Binary Multiplication (13–18)

Q13	Q14	Q15	Q16	Q17	Q18
1111_2	11110_2	100001_2	1000001_2	111100_2	10011010_2

Hexadecimal Addition (19–24)

Q19	Q20	Q21	Q22	Q23	Q24
41_{16}	67_{16}	$C8_{16}$	DF_{16}	$49F_{16}$	1084_{16}

Hexadecimal Subtraction (25–30)

Q25	Q26	Q27	Q28	Q29	Q30
35_{16}	69_{16}	85_{16}	$8B_{16}$	$1ED_{16}$	537_{16}